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Infinite Families of Infinite Series With Integer Sums

Damiano Fulghesu, James A. Sellers, and Courtney K. Taylor

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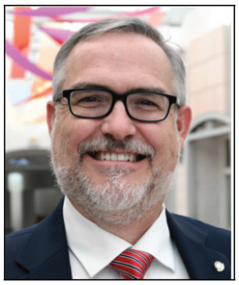

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Those who are familiar with first-year calculus know that one can prove that a series of the form

$$\sum_{n=1}^{\infty} \frac{n^k}{c^n}, \tag{1}$$

where k is a positive integer and c is a real number, converges for any such k and any $c > 1$ using the ratio test. Indeed, numerous calculus textbooks ask students to prove

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convergence of series such as

$$\sum_{n=1}^{\infty} \frac{n^2}{2^n} \quad \text{and} \quad \sum_{n=1}^{\infty} \frac{n^3}{2^n} \tag{2}$$

via the ratio test. Interestingly enough, the two series mentioned in (2) were considered several centuries ago by Jacob Bernoulli in his *Tractatus de Seriebus Infinitis*. And, even more interestingly, Bernoulli went beyond proving that they converge. In fact, he found their exact values:

$$\sum_{n=1}^{\infty} \frac{n^2}{2^n} = 6 \quad \text{and} \quad \sum_{n=1}^{\infty} \frac{n^3}{2^n} = 26. \tag{3}$$

Of course, this then begs the question: Why is it that we don't show students such "exact" results in first-year calculus courses? After all, having examples of infinite series whose sums are **integers** seems about as elegant and elementary as one can get. In response to such a concern, Sellers [**2**] discussed a way to determine the exact values of series in the family (1) using power series manipulations of $\dfrac{1}{1-x}$ via techniques that are often taught in first-year calculus courses.

One modern-day way to prove results such as those mentioned in (3) is to begin with the well-known geometric series,

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$$

which converges for $|x| < 1$, and then apply a tool called the "$x\frac{d}{dx}$ operator" to obtain the desired power series. In this case, one application of this operator, which involves taking the derivative of both sides of an equation with respect to x, and then multiplying by x on both sides, yields

$$\sum_{n=1}^{\infty} nx^n = \frac{x}{(1-x)^2}$$

and a second application gives

$$\sum_{n=1}^{\infty} n^2x^n = \frac{x(1+x)}{(1-x)^3}.$$

All of the results listed above are valid for $|x| < 1$, so we can substitute $x = \frac{1}{2}$ into the above equation to obtain

$$\sum_{n=1}^{\infty} n^2 \left(\frac{1}{2}\right)^n = \sum_{n=1}^{\infty} \frac{n^2}{2^n} = \frac{\frac{1}{2}\left(1+\frac{1}{2}\right)}{\left(1-\frac{1}{2}\right)^3} = 6,$$

after simplification, and this is the first identity in (3). One additional application of the $x\frac{d}{dx}$ operator, followed by the substitution $x = \frac{1}{2}$, yields the second identity in (3).

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With the goal of taking this conversation a bit further, we wish to ask and answer the following question: Is there a way to find an infinite family of infinite series of the form (1) whose sums are **integers**? We provide an answer to this question below, and the path we take allows us to look a bit at what are called the Worpitzky numbers which were recently considered by Vandervelde [**4**].

As we begin the conversation, let's define

$$S(k, x) := \sum_{n=1}^{\infty} n^k x^n$$

for a positive integer k and a real number x. (Note then that the series in (1) are of the form $S\left(k, \frac{1}{c}\right)$ which means, in particular, that we will focus our attention throughout the rest of this work on x such that $0 < x < 1$.)

Initial results via the Worpitzky numbers

We wish to find a representation of $S(k, x)$ which will allow us to see quickly, for particular k and x, whether the sum $S(k, x)$ is an integer. We begin the process of finding such a representation by stating and proving the following elementary theorem which is, arguably, the key idea that we need to prove everything else in this work.

Theorem 1. *For all $j \geq 1$,*

$$x\frac{d}{dx}\left(\frac{1}{(1-x)^j}\right) = -\frac{j}{(1-x)^j} + \frac{j}{(1-x)^{j+1}}.$$

Proof. It's easy to check that both sides of the equation above equal $\dfrac{jx}{(1-x)^{j+1}}$ after a small amount of simplification. ■

Note that Theorem 1 involves the application of the $x\frac{d}{dx}$ operator, something that Sellers [**2**] and Knopf [**1**] have discussed at length. Its use is pivotal to the work below, mainly because $S(k+1, x) = x\dfrac{d}{dx}S(k, x)$, and yet, there's nothing magical about it—it is simply the tool that we use to carry out the necessary power series manipulations.

With Theorem 1 in hand, we can quickly and efficiently write down closed formulas for $S(k, x)$ for many values of k:

$$S(1, x) = \sum_{n=1}^{\infty} nx^n = x\frac{d}{dx}\left(\frac{1}{1-x}\right) = -\frac{1}{1-x} + \frac{1}{(1-x)^2},$$

$$S(2, x) = \sum_{n=1}^{\infty} n^2x^n = x\frac{d}{dx}\left(S(1, x)\right) = \frac{1}{1-x} - \frac{3}{(1-x)^2} + \frac{2}{(1-x)^3},$$

$$\begin{aligned} S(3, x) &= \sum_{n=1}^{\infty} n^3x^n = x\frac{d}{dx}\left(S(2, x)\right) \\ &= -\frac{1}{1-x} + \frac{7}{(1-x)^2} - \frac{12}{(1-x)^3} + \frac{6}{(1-x)^4}, \end{aligned}$$

$k\backslash j$	0	1	2	3	4	5
0	1					
1	1	1				
2	1	3	2			
3	1	7	12	6		
4	1	15	50	60	24	
5	1	31	180	390	360	120

Table 1. Worpitzky numbers $W_{k,j}$.

and so on. One of the obvious questions to ask next is, "Where do those numerators come from?" That is to say, are those numbers already known? Indeed, they are! These are the **Worpitzky numbers** $W_{k,j}$, and they appear in Sloane's OEIS [3] as sequence number A028246. Vandervelde [4] recently discussed these numbers extensively in his very engaging work. In [4], he notes that $W_{k,j}$ counts the number of ways to partition a set of k distinct objects into j ordered subsets. As such, $W_{k,j} = j!\left\{ {k \atop j} \right\}$ where $\left\{ {k \atop j} \right\}$ are the Stirling numbers of the second kind. Vandervelde notes that the integers $W_{k,j}$ satisfy the following relations for all $k, j \geq 0$:

$$W_{k,0} = 1, \quad W_{k,k} = k!, \quad \text{and} \quad W_{k+1,j+1} = (j+2)W_{k,j+1} + (j+1)W_{k,j}.$$

From here, we can easily build a triangle of Worpitzky numbers where the rows are labeled in such a way that the "first" row, which consists of a single 1, is the $k = 0$ row. See Table 1.

The examples $S(1, x)$, $S(2, x)$, and $S(3, x)$ above hint at the pattern given in the following theorem which provides us with our desired closed-form representation for $S(k, x)$:

Theorem 2. *For $k \geq 1$,*

$$S(k, x) = \sum_{j=0}^{k} \frac{(-1)^{k+j} W_{k,j}}{(1-x)^{j+1}}.$$

It's pretty clear from the examples shared above that this is the result. Even so, for completeness' sake, we provide a proof by induction (partly because it's such a straightforward example of proof by induction thanks to the properties of $W_{k,j}$).

Proof. The basis case, $k = 1$, is easy to check.

$$S(1, x) = \sum_{n=1}^{\infty} nx^n = x\frac{d}{dx}\left(\frac{1}{1-x}\right) = \frac{x}{(1-x)^2} = -\frac{1}{1-x} + \frac{1}{(1-x)^2}.$$

Next, we perform the induction step of the proof. Assume

$$S(k, x) = \sum_{j=0}^{k} \frac{(-1)^{k+j} W_{k,j}}{(1-x)^{j+1}}$$

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for some $k \geq 1$. We want to prove that

$$S(k+1, x) = \sum_{j=0}^{k+1} \frac{(-1)^{k+j+1} W_{k+1,j}}{(1-x)^{j+1}}.$$

Note that

$$\begin{aligned}
S(k+1, x) &= \sum_{n=1}^{\infty} n^{k+1} x^n = x \frac{d}{dx}\left(\sum_{n=1}^{\infty} n^k x^n\right) = x \frac{d}{dx}\left(S(k, x)\right) \\
&= x \frac{d}{dx}\left(\sum_{j=0}^{k} \frac{(-1)^{k+j} W_{k,j}}{(1-x)^{j+1}}\right) \quad \text{by the induction hypothesis} \\
&= \sum_{j=0}^{k} (-1)^{k+j} W_{k,j} \left(x \frac{d}{dx}\left(\frac{1}{(1-x)^{j+1}}\right)\right) \\
&= \sum_{j=0}^{k} (-1)^{k+j} W_{k,j} \left(\frac{-(j+1)}{(1-x)^{j+1}}\right) + \sum_{j=0}^{k} (-1)^{k+j} W_{k,j} \left(\frac{(j+1)}{(1-x)^{j+2}}\right)
\end{aligned}$$

using Theorem 1. We now split the two sums above and recombine them intelligently to obtain

$$\begin{aligned}
S(k+1, x) &= (-1)^{k+0} W_{k,0} \left(\frac{-1}{(1-x)}\right) + \sum_{j=1}^{k} (-1)^{k+j} W_{k,j} \left(\frac{-(j+1)}{(1-x)^{j+1}}\right) \\
&\quad + \sum_{j=0}^{k-1} (-1)^{k+j} W_{k,j} \left(\frac{(j+1)}{(1-x)^{j+2}}\right) + (-1)^{k+k} W_{k,k} \left(\frac{(k+1)}{(1-x)^{k+2}}\right) \\
&= \frac{(-1)^{k+1} W_{k+1,0}}{(1-x)} + \sum_{j=1}^{k} \frac{(-1)^{k+j+1}(j+1) W_{k,j}}{(1-x)^{j+1}} \\
&\quad + \sum_{j=1}^{k} \frac{(-1)^{k+j+1}(j) W_{k,j-1}}{(1-x)^{j+1}} + \frac{W_{k+1,k+1}}{(1-x)^{k+2}} \\
&= \frac{(-1)^{k+1} W_{k+1,0}}{(1-x)} + \sum_{j=1}^{k} \frac{(-1)^{k+j+1} W_{k+1,j}}{(1-x)^{j+1}} + \frac{W_{k+1,k+1}}{(1-x)^{k+2}} \\
&= \sum_{j=0}^{k+1} \frac{(-1)^{k+j+1} W_{k+1,j}}{(1-x)^{j+1}}
\end{aligned}$$

using all of the facts about $W_{k,j}$ that were mentioned above. This is the desired result. ■

We now return to our original motivating question: Is there a way to find an infinite family of infinite series of the form (1) whose sums are **integers**? The answer is a resounding YES!

Theorem 3. *Let $r \geq 1$ be an integer. Then, for all $k \geq 1$, $S\left(k, \dfrac{r}{r+1}\right)$ is an integer.*

The proof of this theorem is now straightforward.

Proof. Thanks to Theorem 2, we know that

$$\begin{aligned} S\left(k, \frac{r}{r+1}\right) &= \sum_{j=0}^{k} \frac{(-1)^{k+j} W_{k,j}}{(1-\frac{r}{r+1})^{j+1}} \\ &= \sum_{j=0}^{k} \frac{(-1)^{k+j} W_{k,j}}{(\frac{1}{r+1})^{j+1}} \\ &= \sum_{j=0}^{k} (-1)^{k+j} (r+1)^{j+1} W_{k,j} \end{aligned}$$

and this is clearly an integer since $W_{k,j}$ is an integer for all $k, j \geq 0$. ■

Let's pause for a moment to put this result in context. Remember the two series that we mentioned earlier from Bernoulli's work:

$$\sum_{n=1}^{\infty} \frac{n^2}{2^n} = 6 \text{ and } \sum_{n=1}^{\infty} \frac{n^3}{2^n} = 26.$$

These two results are, respectively, $S\left(2, \dfrac{1}{2}\right) = 6$ and $S\left(3, \dfrac{1}{2}\right) = 26$ where we have used $r = 1$ in both cases. Thanks to the proof of Theorem 3, we see that we can calculate these values as follows:

$$\sum_{n=1}^{\infty} \frac{n^2}{2^n} = 2W_{2,0} - 4W_{2,1} + 8W_{2,2} = 2 - 12 + 16 = 6, \quad \text{and}$$

$$\sum_{n=1}^{\infty} \frac{n^3}{2^n} = -2W_{3,0} + 4W_{3,1} - 8W_{3,2} + 16W_{3,3} = -2 + 28 - 96 + 96 = 26.$$

Of course, this is NOT how Bernoulli calculated these two sums; nevertheless, it's nice to see the same values arise!

Interestingly, even more can be said. In the case when $r = 1$, it turns out that the sums in question are also just sums of the values of the Worpitzky numbers in the corresponding row!

$$\sum_{n=1}^{\infty} \frac{n^2}{2^n} = \sum_{j=0}^{2} W_{2,j} = 1 + 3 + 2 = 6,$$

$$\sum_{n=1}^{\infty} \frac{n^3}{2^n} = \sum_{j=0}^{3} W_{3,j} = 1 + 7 + 12 + 6 = 26,$$

$$\sum_{n=1}^{\infty} \frac{n^4}{2^n} = \sum_{j=0}^{4} W_{4,j} = 1 + 15 + 50 + 60 + 24 = 150,$$

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and so on, thanks to a functional equation that Vandervelde highlights [**4**, (13)].

Returning to Theorem 3, we also see, for example, that

$$S\left(3, \frac{4}{5}\right) = \sum_{n=1}^{\infty} \frac{n^3}{\left(\frac{5}{4}\right)^n} = \sum_{n=1}^{\infty} n^3 \left(\frac{4}{5}\right)^n$$
$$= \sum_{j=0}^{3} (-1)^{j+3} (5)^{j+1} W_{3,j}$$
$$= 2420.$$

Who would have ever thought, a priori, that such a sum would be an integer?!?!

We close this section by highlighting that, since the power series $S(3, x)$ is a continuous function, and since

$$S\left(3, \frac{1}{2}\right) = 26 \text{ and}$$
$$S\left(3, \frac{4}{5}\right) = 2420,$$

there must exist values of x between $\frac{1}{2}$ and $\frac{4}{5}$ such that the series sums to every integer between 26 and 2420. Such an observation related to continuity could lead to numerous avenues of further study.

A deeper dive—the series of Oresme and Swineshead

The results above are extremely satisfying, especially since they provide an infinite family of (rational) values of x which yield infinite series with integer sums. Unfortunately, this is not the full story in the sense that the above does not provide a full characterization of all of the values of k and x such that $S(k, x)$ is an integer. In order to take this further, we now focus our attention on the simplest of the series of this form, namely $S(1, x) = \sum_{n=1}^{\infty} nx^n$. This particular set of series is extremely important for a few different reasons. Historians of mathematics often note that $S(1, \frac{1}{2}) = \sum_{n=1}^{\infty} n \left(\frac{1}{2}\right)^n = 2$ was computed by the contemporaries Oresme in France and Swineshead in England in the mid 14th century. For our sake, this series is of significant importance because of the simplicity of its closed-form representation as

$$S(1, x) = \sum_{n=1}^{\infty} nx^n = -\frac{1}{1-x} + \frac{1}{(1-x)^2}, \tag{4}$$

which we highlighted earlier.

If we utilize the change of variables $y = \dfrac{1}{1-x}$, then we see that

$$S(1, x) = \sum_{n=1}^{\infty} nx^n = -\frac{1}{1-x} + \frac{1}{(1-x)^2} = -y + y^2. \tag{5}$$

We can now select a positive integer L and ask whether we can find an x such that $L = \displaystyle\sum_{n=1}^{\infty} nx^n$. Clearly, this is equivalent to solving $y^2 - y - L = 0$ for y. Thanks

to the quadratic formula, we can solve for y and obtain $y = \dfrac{1+\sqrt{4L+1}}{2}$, and so $x = 1 - \dfrac{2}{1+\sqrt{4L+1}}$.

For positive L, it's clear that $0 < x < 1$, and so putting this all together, we've established

$$L = \sum_{n=1}^{\infty} n \left(1 - \frac{2}{1+\sqrt{4L+1}}\right)^n. \tag{6}$$

We note that we can "recover" our previous results from (6), at least for $k = 1$. We begin by asking the question, "What values of L will make $4L + 1$ a perfect square?" That's not so hard; $L = r(r+1)$ for some integer r will do the trick because

$$4L + 1 = 4r(r+1) + 1 = (2r+1)^2.$$

If we then substitute such a value for L into $x = 1 - \dfrac{2}{1+\sqrt{4L+1}}$, we have

$$x = 1 - \frac{2}{1+\sqrt{4L+1}} = 1 - \frac{2}{1+\sqrt{4r(r+1)+1}} = \frac{r}{r+1}$$

after simplification. This is reminiscent of Theorem 3, at least in the case $k = 1$.

Of course, we can use (6) to obtain **any** desired positive integer L as the sum of a convergent series if we use the appropriate (sometimes irrational) x. In particular, the case $L = 1$ is particularly fun in this context given the appearance of the golden ratio φ:

$$1 = \sum_{n=1}^{\infty} n \left(1 - \frac{2}{1+\sqrt{5}}\right)^n = \sum_{n=1}^{\infty} n \left(1 - \frac{1}{\varphi}\right)^n = \sum_{n=1}^{\infty} n\,(2-\varphi)^n.$$

As an aside, it is intriguing to note that

$$1 - \frac{2}{1+\sqrt{5}} = 1 - \frac{1}{\varphi} = \frac{\varphi - 1}{\varphi},$$

which means the sum above is of the form $S\left(1, \dfrac{r}{r+1}\right)$ with $r = \varphi - 1$.

Of course, there's nothing mysterious about the appearance of the golden ratio here since finding the value of x for which $S(1, x) = 1$ is equivalent to solving the equation $y^2 - y - 1 = 0$ for y. Interestingly enough, the golden ratio appears to provide lots of other examples of series of the form $S(k, x)$ whose sums are integers. Numerical evidence indicates the following:

$$\sum_{n=1}^{\infty} n^3\,(2-\varphi)^n = 7, \qquad \sum_{n=1}^{\infty} n^5\,(2-\varphi)^n = 151, \qquad \sum_{n=1}^{\infty} n^7\,(2-\varphi)^n = 6847.$$

Note that these numbers, 1, 7, 151, and 6847, are the first few values of the sequence A100868 in Sloane's OEIS [**3**], and they are connected to that sequence by the fact that $2 - \varphi = \varphi^2$. We will show that the above sums are equal to these integers in the next section.

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As we close this section, we note that, if the method used above to analyze $S(1, x)$ is attempted on $S(k, x)$ for $k > 1$, then the polynomial in the variable y that arises, as in (5), will have higher degree. For $k = 2$ and $k = 3$, the corresponding polynomials in y will have degree 3 and 4, respectively, so in theory, we could find y (and, therefore, x). Of course, for $k > 3$, the corresponding equations are not solvable by radicals except for special cases.

One more cool pattern

The set of results at the end of the previous section, where the sums equal 1, 7, 151, and 6847 respectively, begs another question: Are those particular sums "unique", or are they part of a much larger phenomenon related to series with integer sums? It turns out that the latter is true—anytime $S(1, x)$ is an integer, it is the case that $S(k, x)$ is an integer for any odd k as well! Let's walk through a proof of this claim.

Notice that for an odd number k, $S(k, x)$ can be obtained from $S(1, x)$ by applying the $x\frac{d}{dx}$ operator an even number of times. So let's consider what happens when we apply the operator $x\frac{d}{dx}x\frac{d}{dx}$ to powers of the series $S(1, x)$.

First, keeping in mind that $x\frac{d}{dx}S(1, x) = S(2, x)$, we use the chain rule to obtain

$$x\frac{d}{dx}S(1, x)^n = nS(1, x)^{n-1}x\frac{d}{dx}S(1, x) = nS(1, x)^{n-1}S(2, x).$$

When we apply the $x\frac{d}{dx}$ operator a second time, we see that

$$x\frac{d}{dx}\left(x\frac{d}{dx}S(1, x)^n\right) = n(n-1)S(1, x)^{n-2}S(2, x)^2 + nS(1, x)^{n-1}S(3, x).$$

Thanks to Theorem 2, we know that

$$S(2, x)^2 = \left(\frac{1}{1-x} - \frac{3}{(1-x)^2} + \frac{2}{(1-x)^3}\right)^2 \text{ and}$$

$$S(3, x) = -\frac{1}{1-x} + \frac{7}{(1-x)^2} - \frac{12}{(1-x)^3} + \frac{6}{(1-x)^4}.$$

Using algebraic simplification, we can then verify the following identities:

$$S(2, x)^2 = 4S(1, x)^3 + S(1, x)^2 \text{ and}$$

$$S(3, x) = 6S(1, x)^2 + S(1, x).$$

And so, with a bit of substitution, it is apparent that the $x\dfrac{d}{dx}x\dfrac{d}{dx}$ operator applied to $S(1, x)^n$ produces a sum of powers of $S(1, x)$ that are multiplied by integers:

$$x\frac{d}{dx}\left(x\frac{d}{dx}S(1, x)^n\right) = (4n^2 + 2n)S(1, x)^{n+1} + n^2S(1, x)^n. \tag{7}$$

With Equation (7) in hand, a straightforward proof by induction shows that, for any odd $k = 2l - 1$,

$$S(k, x) = \sum_{j=1}^{l} a_j S(1, x)^j$$

with a_j integers. And a direct consequence of this result is the following theorem:

Theorem 4. *For any odd k, if $S(1, x)$ is an integer, then $S(k, x)$ is an integer divisible by $S(1, x)$.*

Using the above work, we can show

$$S(3, x) = 6S(1, x)^2 + S(1, x),$$
$$S(5, x) = 120S(1, x)^3 + 30S(1, x)^2 + S(1, x), \quad \text{and}$$
$$S(7, x) = 5040S(1, x)^4 + 1680S(1, x)^3 + 126S(1, x)^2 + S(1, x).$$

So, as we saw above, since $S(1, 2 - \varphi) = 1$, we know that

$$S(3, 2 - \varphi) = 6 + 1 = 7,$$
$$S(5, 2 - \varphi) = 120 + 30 + 1 = 151, \quad \text{and}$$
$$S(7, 2 - \varphi) = 5040 + 1680 + 126 + 1 = 6847.$$

Theorem 4 begs a fairly obvious question—could a similar result also hold for $S(k, x)$ where k is **even**? It turns out that a result does exist, although it's not as neat as the statement of the above theorem. It turns out that, if $S(1, x)$ and $S(2, x)$ are both integers, then for all positive integers j, $S(2j, x)$ is an integer divisible by both $S(1, x)$ and $S(2, x)$. We leave the details to the interested reader.

Closing thoughts

As we close, let's take a moment to highlight what we've done and also consider potential avenues for future work.

First, we've shown that, for all integers $k \geq 1$ and all integers $r \geq 1$, the infinite series of the form

$$\sum_{n=1}^{\infty} \frac{n^k}{\left(\frac{r+1}{r}\right)^n}$$

sums to an integer, and we've given a closed-form formula for what each such series equals. We've also noted that, for any positive integer L that we wish, we can find a nontrivial series (in particular, a series in the family where $k = 1$) whose sum equals L. Lastly, we've indicated an infinite family of values of x, other than those of the form $\frac{r}{r+1}$, which still yield integer sums (particularly in the case where $k = 1$).

Unfortunately, we've fallen short of providing a characterization of all of the values of c which guarantee that the series

$$\sum_{n=1}^{\infty} \frac{n^k}{c^n}$$

sums to an integer. The interested reader may wish to take such an analysis further; if so, we recommend Vandervelde's article [**4**] once again as he provides some interesting results on the divisibility properties of $W_{k,j}$. We also encourage the interested reader to consider the change of variable $y = \frac{1}{1-x}$, which was the focus of the previous section, in the case of the family of series $S(k, x)$ for specific $k > 1$.

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Of course, such characterization work may be extremely difficult. Notice, in light of the previous section, that there can be odd k for which $S(k, x)$ is an integer while $S(1, x)$ is not. For example, using (4) we know

$$S\left(1, 1 - \frac{2}{1+\sqrt{3}}\right) = \frac{1}{2}.$$

However,

$$\begin{aligned} S\left(3, 1 - \frac{2}{1+\sqrt{3}}\right) &= 6S\left(1, 1 - \frac{2}{1+\sqrt{3}}\right)^2 + S\left(1, 1 - \frac{2}{1+\sqrt{3}}\right) \\ &= 6\left(\frac{1}{2}\right)^2 + \frac{1}{2} = 2. \end{aligned}$$

Ultimately, we hope that this work provides sufficient motivation to encourage our colleagues to show these kinds of integer sum results to their first-year calculus students!

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Summary. Motivated by a desire to show first-year calculus students examples of infinite series whose sums are relatively straightforward to find, we demonstrate a technique to calculate such sums using tools often demonstrated to calculus students. We then use these results, which include an explicit formula for the sums in question, to prove the existence of an infinite family of infinite series whose sums are integers.

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